Software Project Management

Post Mortem Report

For

AuralAid: Adaptive AI Hearing with Contextual Understanding

Prepared by

Section Name: B

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Post Mortem Report

The post mortem for "AuralAid: Adaptive AI Hearing with Contextual Understanding" reflects on the development journey of an intelligent assistive hearing system integrating AI, embedded systems, and mobile application technologies. The project aimed to provide real time sound classification, speech recognition, contextual alerts, and user assistance for individuals with hearing impairments. This evaluation highlights initial expectations, milestones, achievements, challenges, stakeholder feedback, risks, and lessons learned throughout the project lifecycle. The purpose of this post-mortem is to document insights and improvements for future AI and IoT-based healthcare projects.

Project Name	AuralAid: Adaptive AI Hearing with Contextual Understanding
Meeting Date/Time	2026-5-24, 10:00 AM
Meeting Facilitator	Junaid Ali Khan
Meeting Attendees	Hamza Moeen, Ayesha Nazir

Initial Expectations
At the initiation of the project, the goal was to develop a smart AI powered hearing assistance system capable of real time sound detection, speech recognition, noise filtering, and contextual awareness. The system was expected to integrate embedded hardware (ESP32, microphones, sensors), AI models for audio processing, and a Flutter-based mobile application for visualization and user interaction. The project was planned to be completed within an academic timeline with deliverables including hardware prototype, AI model integration, mobile application interface, and system testing. Key deliverables included: • Hardware prototype development • AI model integration for sound and speech recognition • Mobile application development • System integration and testing • Final system deployment and documentation

Milestones

1. Project Title Defense
2. SRS and Proposal Approval
3. System Architecture Design
4. Hardware Component Selection and Integration
5. AI Model Selection and Testing
6. Embedded System Programming
7. Mobile Application UI Design
8. AI and Hardware Integration
9. Real Time Audio Processing Implementation
10. Testing Phase Completion
11. Bug Fixes and Optimization
12. Final System Demonstration and Submission

Achievements and Obstacles

Achievements:

- Successfully integrated MEMS microphones with ESP32 for real time audio capture.
- Implemented AI based sound classification and speech recognition models.
- Developed a Flutter based mobile application for real time alerts and visualization.
- Designed a modular system architecture enabling scalability and upgrades.
- Implemented noise detection, vibration alerts, and emergency sound notifications.
- Achieved successful communication between hardware and mobile application via wireless protocols.

Obstacles:

- Difficulty in deploying AI models on low resource embedded devices.
- Delays due to hardware compatibility and power management issues.
- Limited dataset availability for training and testing AI models.
- Challenges in real time processing and latency optimization.
- Integration complexity between AI, hardware, and mobile application layers.

Stakeholder Input
• **Advisor**: Appreciated the innovative healthcare based assistive technology concept and interdisciplinary integration of AI, IoT, and mobile development. Recommended focusing on system optimization, reliability, and real world usability. • **Test Players**: Valued real time sound detection, vibration alerts, and caption features. Suggested improving speech recognition accuracy, noise filtering, and response time in noisy environments. • **Team Members**: Highlighted strong collaboration across hardware, AI, and app development modules. Noted challenges in time management, synchronization between modules, and workload balancing. • **Faculty Panel**: Recognized the technical depth and real world impact for hearing-impaired users. Appreciated the integration of AI models with embedded systems and mobile platforms. • **Peer Reviewers**: Commended the innovative problem solving approach and system architecture. Suggested improvements in UI clarity and user onboarding experience in the mobile app. • **Academic Review Board**: Applauded the project's societal value and technical execution as a final year project. Recommended planning scalability and future healthcare integrations.

Roadblocks, Risks and Barriers

- **AI Model Optimization Challenges:** Difficulty in running speech recognition and emotion detection models efficiently on low resource embedded devices.
- **Hardware Integration Issues:** Synchronization problems between MEMS microphones, ESP32-S3, and mobile app communication.
- **Bluetooth Communication Instability:** Data transmission delays and occasional disconnections between the wearable device and mobile application.
- **Unforeseen Technical Issues:** Debugging DSP processing, AI inference delays, and real-time streaming complexities increased development time.
- **Feasibility of Real-Time Processing:** Challenges in achieving low latency audio processing while maintaining accuracy.
- **Resource Constraints:** Limited access to specialized hardware components and testing devices.
- **Limited Testing Time:** Restricted real-world user testing affected validation of system reliability and performance.

Additional Resources

- **Advanced AI Computing Resources:** Access to GPU servers or AI accelerators for faster model training and optimization.
- **Professional Healthcare Consultation:** Expert input from audiologists and hearing specialists for system validation.
- **Dedicated QA Testing Team:** For extensive real world scenario testing and system reliability verification.
- **High-End Embedded Hardware:** Advanced microcontrollers and DSP processors for better performance.
- **Extended User Testing Period:** More time for testing with hearing impaired users for usability refinement.
- **Cloud AI Infrastructure:** For scalable AI processing and future system expansion.

Root Causes
The core challenges originated from the complex integration of AI, embedded systems, real-time signal processing, and mobile application development within a limited timeframe. Managing technical complexity, hardware constraints, AI optimization, limited resources, and parallel module development created coordination and performance challenges throughout the project lifecycle.

Key Takeaways
Effective project management, modular system design, early feasibility testing, and realistic technical planning are essential when developing AI based healthcare systems. Early integration testing, proper resource planning, and continuous validation significantly improve system stability and project success.

Action Items
• Optimize AI models for low resource devices • Improve Bluetooth and data transmission stability • Enhance noise filtering and DSP processing • Strengthen hardware software synchronization • Improve UI/UX for accessibility • Expand modular system architecture • Implement structured testing framework • Improve documentation and system manuals • Schedule regular technical integration reviews • Plan scalability and future healthcare features

Next Steps
• **Share Post-Mortem Report:** Distribute the report to all stakeholders and team members. • **Assign Action Items:** Allocate tasks to AI, hardware, and app development teams. • **Monitor Progress:** Track completion using project management tools. • **Schedule Follow-Up Meetings:** Review implementation progress regularly. • **Prepare for Future Updates:** Plan next development phase and feature upgrades. • **Document Lessons Learned:** Maintain a structured knowledge base for future projects. • **User Feedback Integration:** Begin structured real user feedback collection. • **System Optimization Phase:** Focus on performance, reliability, and scalability.